A Network Approach to Geopolitics

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Geopolitics in an interconnected world

- Geopolitics: a fast-growing literature in economics ("geoeconomics")
  - theory: a single hegemon, or a hegemon vs. a rising power
  - empirics: one channel at a time (trade, sanctions, finance, aid, alliances)
- But geopolitics is inherently a network phenomenon: many countries and many channels at once
  - Cold War: the West and the Soviets fought for influence over African and Southeast Asian countries
  - Russia–Ukraine war: coordination problem across Europe & North America, over sanctions, arms, energy
  - Trump & Greenland: within days, a coalition of eight European partners issued a unified response
- Common thread: countries shape foreign policy by influencing third parties

This paper: a network approach to geopolitics

Model

- Countries take geopolitical actions, balancing own interests and alignment with network neighbors
- Before acting, countries can exert costly influence on other countries' incentives
- Influence propagates through the network; bilateral influence has cascading effects

Empirics

- Treat geopolitical actions as equilibrium outcomes; recover the network via revealed preference
  - compare what countries want with what they do: no need to pick a channel
- Validate with trade, cultural, diplomatic, and security data; perform geopolitical counterfactuals

Model setup

- N countries connected by a geopolitical network $\boldsymbol{G}$
  - matrix $\boldsymbol{G} \in \mathbb{R}_{\geq 0}^{N \times N}$ with row sums $1 - \alpha$; symmetric (relaxed later)
  - a reduced-form summary of long-standing relations: trade, security, diplomatic ties and their interactions
- Each country i has domestic interest $\theta_i \in \mathbb{R}$ and chooses a geopolitical action $a_i \in \mathbb{R}$
  - interpret $\boldsymbol{\theta}$ as domestic interests over specific issues
  - e.g., each country's interest in restricting China's access to advanced semiconductors
- Each country i can exert costly influence over any other country k to change k's incentives
  - e.g., US pressure on the Netherlands and Japan to halt chip-equipment sales to China
- Perfect information. One issue $\boldsymbol{\theta}$ for now; the data have many issues, each a separate draw of $\boldsymbol{\theta}$

Model without foreign influence

- Without foreign influence, countries play a geopolitical beauty-contest game:

$$U_i = \max_{a_i} - \left[\alpha \left(a_i - \theta_i \right)^2 + \sum_j G_{ij} \left(a_i - a_j \right)^2 \right]$$

- Equilibrium:

$$a_i = \alpha \theta_i + \sum_j G_{ij} a_j,$$

or, in matrix form,

$$\begin{aligned} \boldsymbol{a} &= \alpha \left(\boldsymbol{I} + \boldsymbol{G} + \boldsymbol{G}^2 + \cdots \right) \boldsymbol{\theta} \\ &= \alpha \left(\boldsymbol{I} - \boldsymbol{G} \right)^{-1} \boldsymbol{\theta} \end{aligned}$$

Model with foreign influence

- Add influence ω_{im} (first index is the receiver of influence, second index is the sender):

$$U_i = -\left[\alpha\left(a_i-\theta_i\right)^2+\sum_j G_{ij}\left(a_i-a_j\right)^2\right] \underbrace{+2\sum_m \omega_{im}\left(a_i-\theta_i\right)}_{\equiv\text{change in incentive due to foreign influence}} \underbrace{-\kappa\sum_k \omega_{ki}^2}_{\substack{\text{cost of exerting influence}\\ \text{(scaled by } \kappa\text{)}}}$$

- Ex-ante: each country i chooses $\{\omega_{ki}\}_k$
- Ex-post: each country i chooses a_i, taking $\mathbf{\Omega}$ and the other countries' actions as given

Model with foreign influence: equilibrium

- Ex-post equilibrium $\boldsymbol{a}$, given the matrix $\boldsymbol{\Omega}$ of bilateral influence:

$$a_i = \underbrace{\alpha\theta_i + \sum_j G_{ij}a_j}_{\substack{\text{beauty-contest} \\ \text{best response}}} + \underbrace{\sum_m \omega_{im}}_{\substack{\text{net influence} \\ \text{received}}} \qquad \Longleftrightarrow \qquad \boldsymbol{a} = \alpha\left(\boldsymbol{I} - \boldsymbol{G}\right)^{-1}\left(\boldsymbol{\theta} + \boldsymbol{\Omega}\mathbf{1}/\alpha\right)$$

  - implication: influencing country k has global effects through geopolitical coordination:

$$\frac{\partial a_j}{\partial \omega_{ki}} = \left[\left(\boldsymbol{I} - \boldsymbol{G}\right)^{-1}\right]_{jk}$$

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- Ex-ante equilibrium $\boldsymbol{\Omega}$ solves $\frac{\mathrm{d}\, U_i(\boldsymbol{\Omega}, \boldsymbol{a}(\boldsymbol{\Omega}))}{\mathrm{d}\omega_{ki}} = 0$, implying

$$\kappa \omega_{ki} = \sum_j G_{ij} \left(a_i - a_j\right) \left(\boldsymbol{I} - \boldsymbol{G}\right)^{-1}_{jk}$$

  - Country i gains more from influencing k when
    - its neighbors j act out of line with i (via $\boldsymbol{G}_{ij}\left(a_i - a_j\right)$)
    - and those neighbors listen to k (via $\left(\boldsymbol{I} - \boldsymbol{G}\right)^{-1}_{jk}$)

Proposition. The equilibrium actions $\boldsymbol{a}$ follow

$$\boldsymbol{a} = \mathcal{L}(\boldsymbol{G})\,\boldsymbol{\theta}, \qquad \mathcal{L}(\boldsymbol{G}) \equiv (\kappa\alpha - 1)\left(\kappa(\boldsymbol{I} - \boldsymbol{G})^2 - \alpha\boldsymbol{I}\right)^{-1}(\boldsymbol{I} - \boldsymbol{G}).$$

Equilibrium net influence follows $\boldsymbol{\Omega 1} = -\left(\kappa(\boldsymbol{I} - \boldsymbol{G})^2 - \alpha\boldsymbol{I}\right)^{-1}\left((\boldsymbol{I} - \boldsymbol{G})^2 - \alpha^2\boldsymbol{I}\right)\boldsymbol{\theta}$.

- Shut down influence $\Rightarrow$ pure beauty contest: as $\kappa \nearrow \infty$, $\boldsymbol{a} \to \alpha(\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{\theta}$
- Cheap influence homogenizes the world: as $\kappa \searrow 1/\alpha$, all actions converge to the global average $\bar{\theta}$
- Equilibrium properties: unique; each a_i is a convex combination of $\boldsymbol{\theta}$; translation invariant to $\boldsymbol{\theta}$
  - average action = average domestic interest; geopolitical influence nets out to zero
- Equilibrium implication: "power" emerges from $\boldsymbol{G}$ and $\boldsymbol{\theta}$ jointly
  - same network, different issues $\Longrightarrow$ different pivotal countries

Which disagreements lead to large divergence in actions?

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- Eigendecomposition: $\boldsymbol{G} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}'$ with orthonormal basis $\boldsymbol{U}$.
- Each eigenvector $\boldsymbol{u}_\ell$ is a distinct latent geopolitical cleavage on the network
  - countries with $\boldsymbol{u}_\ell(i) \gg 0$ and those with $\boldsymbol{u}_\ell(i) \ll 0$ oppose each other; $\boldsymbol{u}_\ell(i) \approx 0$ on neither side

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$$\boldsymbol{\theta} = \sum_{\ell=1}^{N} \widetilde{\theta}_\ell \boldsymbol{u}_\ell$$

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Proposition. $\boldsymbol{a} = \sum_{\ell=1}^{N} L(\lambda_\ell) \times \widetilde{\theta}_\ell \boldsymbol{u}_\ell$, where $L(\lambda) = \frac{(1-\lambda)(\kappa\alpha-1)}{\kappa(1-\lambda)^2-\alpha}$.

- Interpretation: equilibrium actions amplify or dampen each cleavage by $L(\lambda_\ell)$
  - $L'(\lambda_\ell) > 0$: cleavages with higher eigenvalue $\lambda_\ell \Rightarrow$ more divergence in actions

Why? Eigenvalues measure agreement among neighbors

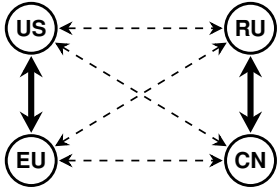
- Consider the network disagreement measure $\mathcal{S}(\boldsymbol{\theta}) \equiv \frac{1}{2}\sum_{i,j} G_{ij}(\theta_i - \theta_j)^2$, which captures how much neighbors' interests differ on average.

Lemma. The network disagreement of an eigenvector $\boldsymbol{u}_\ell$ is $\mathcal{S}(\boldsymbol{u}_\ell) = 1 - \alpha - \lambda_\ell$. Moreover,

$$\boldsymbol{u}_1 = \underset{\|\boldsymbol{\theta}\|=1}{\arg\min}\, \mathcal{S}(\boldsymbol{\theta}) \propto \mathbf{1}, \qquad \boldsymbol{u}_\ell = \underset{\boldsymbol{\theta}\perp\{\boldsymbol{u}_k\}_{k<\ell}, \|\boldsymbol{\theta}\|=1}{\arg\min}\, \mathcal{S}(\boldsymbol{\theta}) \text{ for all } \ell > 1.$$

- high-λ eigenvectors: cleavages where neighbors agree, blocs disagree $\Rightarrow$ amplified (echo chamber)
- low-λ eigenvectors: cleavages where neighbors disagree $\Rightarrow$ dampened by coordination

Example: disagreement across blocs (US–EU vs. Russia–China)



$$\boldsymbol{G} = \begin{array}{c} \\ \text{US} \\ \text{EU} \\ \text{RU} \\ \text{CN} \end{array} \begin{array}{c} \begin{array}{cccc} \text{US} & \text{EU} & \text{RU} & \text{CN} \end{array} \\ \begin{bmatrix} 0 & .7 & .1 & .1 \\ .7 & 0 & .1 & .1 \\ .1 & .1 & 0 & .7 \\ .1 & .1 & .7 & 0 \end{bmatrix} \end{array}$$

- Eigenvectors and eigenvalues:

$$\boldsymbol{u}_\ell \propto \begin{array}{c} \\ \text{US} \\ \text{EU} \\ \text{RU} \\ \text{CN} \end{array} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

$$\lambda_\ell = \quad 0.9, \quad 0.5, \quad -0.7, \quad -0.7$$

Recovering the unobserved network $\boldsymbol{G}$ when $\boldsymbol{\Sigma}_\theta$ is invertible

- Countries act on many international issues $\boldsymbol{\theta} \sim N(\mathbf{0}, \boldsymbol{\Sigma}_\theta)$

$$\boldsymbol{a} = \mathcal{L}(\boldsymbol{G})\,\boldsymbol{\theta} \qquad \Longrightarrow \qquad \boldsymbol{\Sigma}_a = \mathcal{L}(\boldsymbol{G})\,\boldsymbol{\Sigma}_\theta \mathcal{L}(\boldsymbol{G})'$$

- When $\boldsymbol{\Sigma}_\theta$ is invertible, the network is recovered by

$$\mathcal{L}(\boldsymbol{G}) = \boldsymbol{\Sigma}_{\boldsymbol{\theta}}^{-1/2} \left(\boldsymbol{\Sigma}_{\boldsymbol{\theta}}^{1/2} \boldsymbol{\Sigma}_{\boldsymbol{a}} \boldsymbol{\Sigma}_{\boldsymbol{\theta}}^{1/2} \right)^{1/2} \boldsymbol{\Sigma}_{\boldsymbol{\theta}}^{-1/2}$$

  - if $\boldsymbol{\Sigma}_\theta$ and $\boldsymbol{\Sigma}_a$ commute (i.e., share principal components), this simplifies to

$$\mathcal{L}(\boldsymbol{G}) = \left(\boldsymbol{\Sigma}_{\boldsymbol{a}} \boldsymbol{\Sigma}_{\boldsymbol{\theta}}^{-1} \right)^{1/2}$$

Recovering the unobserved network $\boldsymbol{G}$ when $\boldsymbol{\Sigma}_\theta$ is not invertible

- $\boldsymbol{\Sigma}_\theta$ not invertible in the data: many countries, but interests vary along only a few cleavages
- Can still identify $\boldsymbol{G}$ along those cleavages under one assumption:

Assumption *(Graph Stationarity)*. $\boldsymbol{\Sigma}_\theta$ shares the eigenvectors of $\boldsymbol{G}$.

- Interests vary along the same cleavages as the network (allies, shared media, ideology, history)
  - empirically, assumption equivalent to $\boldsymbol{\Sigma}_\theta$ and $\boldsymbol{\Sigma}_a$ sharing principal components

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  - empirically, assumption equivalent to $\boldsymbol{\Sigma}_\theta$ and $\boldsymbol{\Sigma}_a$ sharing principal components
- Then $\boldsymbol{\Sigma}_\theta$ and $\boldsymbol{\Sigma}_a$ are both diagonal in the eigenbasis, and each cleavage is scaled by $L(\lambda_\ell)$:

$$\sigma^2_{a,\ell} = L(\lambda_\ell)^2 \, \sigma^2_{\theta,\ell}$$

Proposition (*Recoverability*). If $\sigma^2_{\theta,\ell} > 0$: $\lambda_\ell = L^{-1}\left(\frac{\sigma_{a,\ell}}{\sigma_{\theta,\ell}}\right)$; collecting cleavages, $\boldsymbol{G} = \boldsymbol{U}\,\text{Diag}(\lambda_\ell)\,\boldsymbol{U}'$.

- Cleavages with no variation in interests ($\sigma^2_{\theta,\ell} = 0$) aren't identified, but never move actions either

Asymmetric network G: structural importance

- So far G is symmetric, but geopolitics isn't: the US matters more to Peru than Peru does to the US
- One new object: country i's structural importance π_i governs all asymmetry:

$$G_{ij}/G_{ji} = \pi_j/\pi_i$$

  - high-π countries: disproportionate impact on neighbors, less sensitive to neighbors' actions
- Influence costs scale the same way: i pays $\kappa\left(\pi_k/\pi_i\right)\omega_{ki}^2$ to influence k
  - expensive to influence high-π targets; expensive to project influence from low-π countries
- All results go through in π-weighted coordinates

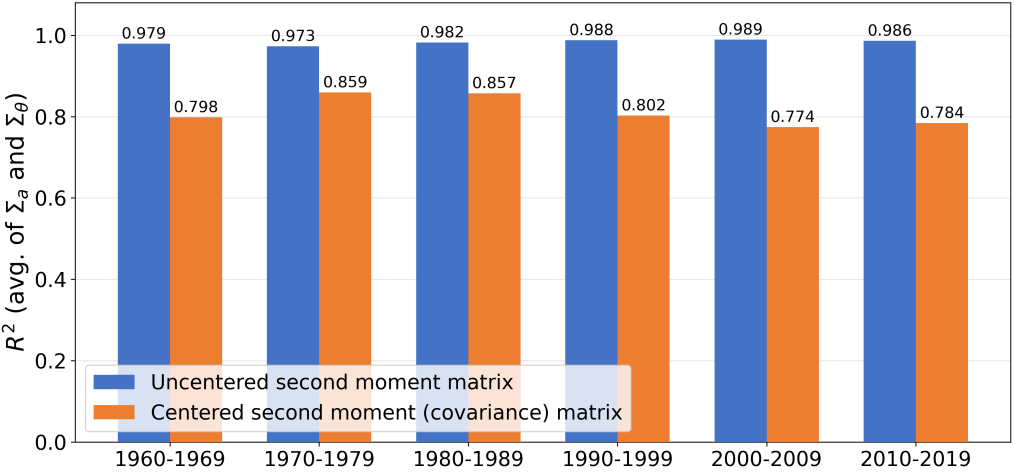
Data

- UN General Assembly data on voting ideal points (Bailey, Strezhnev & Voeten 2017)
  - votes across six topics (human rights, Middle East, nuclear, disarmament, colonialism, economics)
  - capture geopolitical actions taken after international coordination and influence
  - calculate Σ_a as covariance of i and j's actions across topics p within a decade
- V-Dem dataset on various domestic political dimensions (eg freedom of press, gender equality)
  - proxies for characteristics that shape domestic interests before international coordination and influence
  - calculate Σ_θ as covariance of i and j's scores across issues m within a decade
- Use $\pi_i \equiv \text{GDP}_i^\beta$ with $\beta = 0.75$, calibrated so model-implied outward/inward influence vs. GDP matches the data (Formal Bilateral Influence Capacity, FBIC)
  - state-sanctioned, publicly acknowledged dependence (via diplomacy, arms transfers, and trade)

Empirics

- Validation
- Descriptives
- Counterfactuals

The top five components explain most of the variation in second moments



Validation: G correlates with political distance, gravity, and cultural ties

	Outcome: G_{ijd}					
	$\log(\text{Distance}_{ij})$	$\lvert\text{Polity}_{id} - \text{Polity}_{jd}\rvert$	$\log(\text{Trade flow}_{ijd})$	Common religion$_{ij}$	Common language$_{ij}$	Common colonizer$_{ij}$
	(1)	(2)	(3)	(4)	(5)	(6)
x_{ijd}	-0.0879***	-0.0937***	0.0363***	0.0416***	0.0927***	0.0567***
	(0.00679)	(0.00794)	(0.00882)	(0.00452)	(0.0199)	(0.00976)
Observations	112144	104846	61582	110218	112144	112144
Sender × decade FE:	✓	✓	✓	✓	✓	✓
Receiver × decade FE:	✓	✓	✓	✓	✓	✓

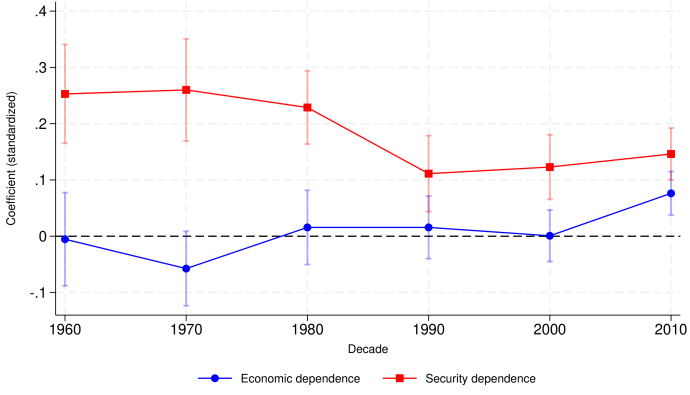
Validation: model-implied influence predicts observable diplomacy

- Recover influence by inverting UN votes; average by country pair and year
- Regress leader visits between two countries on their implied influence (OLS)
- Two countries shifting each other's UN vote by one step (e.g., abstain→yes) on 1% of roll calls ⇒ 0.08–0.1 more leader visits per year (0.04–0.05 with bilateral controls)
  - Bilateral controls: log distance; log bilateral trade; absolute Polity-score gap; common language; common religion; common colonizer

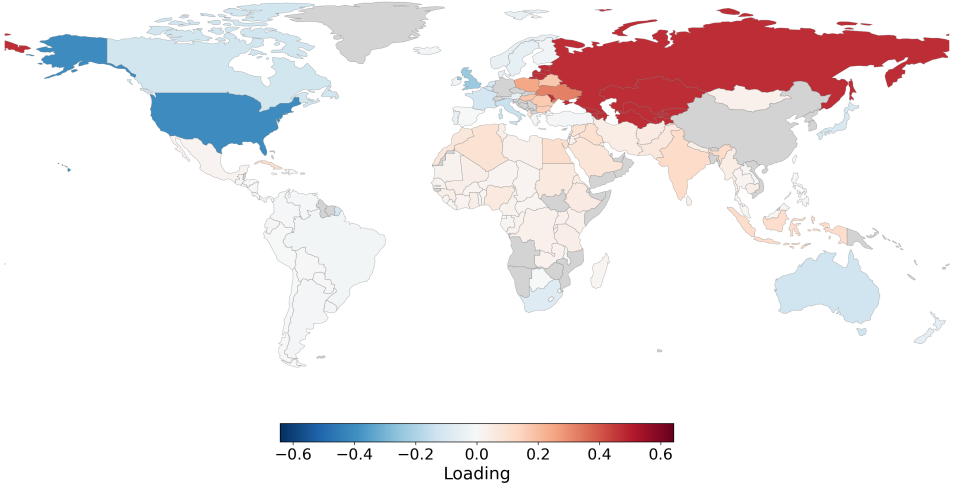
	Outcome: state visits$_{(ij),t}$					
	Annual		3-year bins		5-year bins	
	(1)	(2)	(3)	(4)	(5)	(6)
Influence, $\overline{\omega}_{(ij)t}$	9.2180***	4.5963**	31.3730***	15.9142**	42.0385***	18.5346*
	(2.9034)	(1.9241)	(10.0312)	(7.0897)	(13.5018)	(9.4734)
Shared country-period FE	✓	✓	✓	✓	✓	✓
Bilateral controls	✗	✓	✗	✓	✗	✓
Observations	24,572	24,572	16,809	16,809	13,468	13,468

Economic vs. security dependence over time

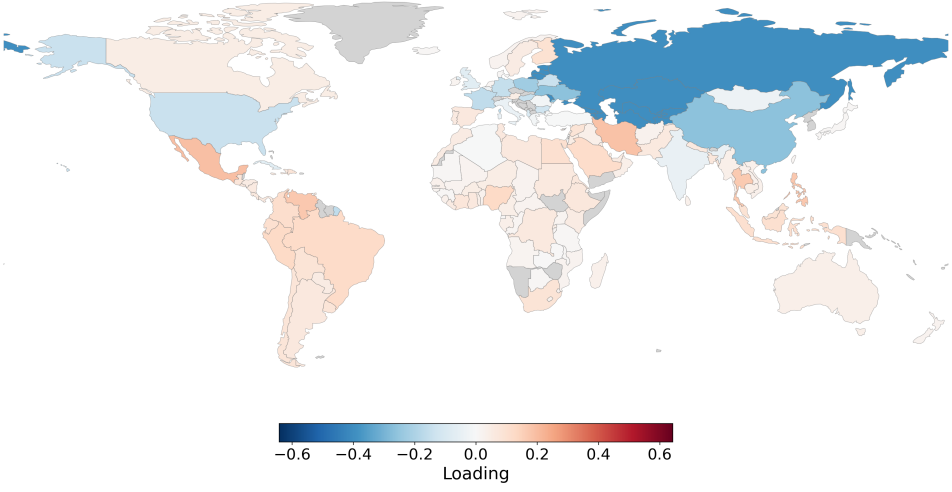
- Regress G_{ijd} on FBIC economic and security dependence by decade, with sender and receiver FE
- Security dependence dominates, but the gap narrows over time and is insignificant by the 2010s



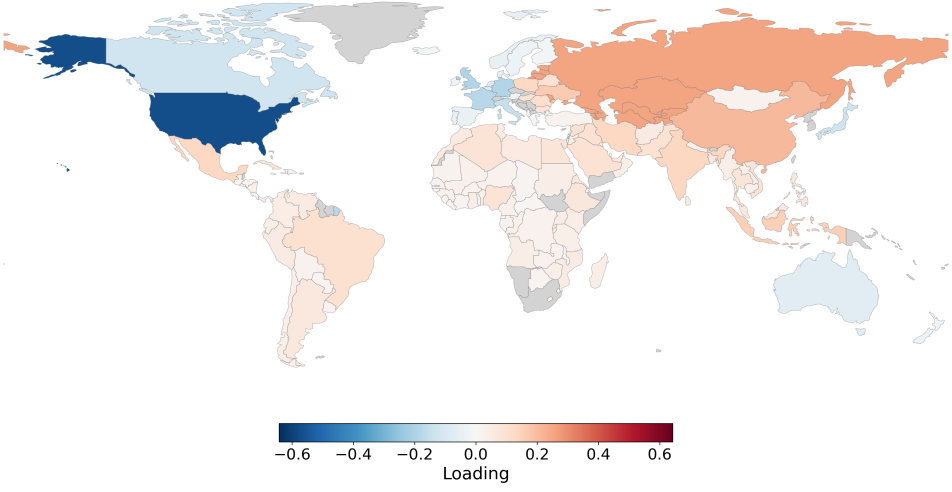
The dominant cleavage: 1960s



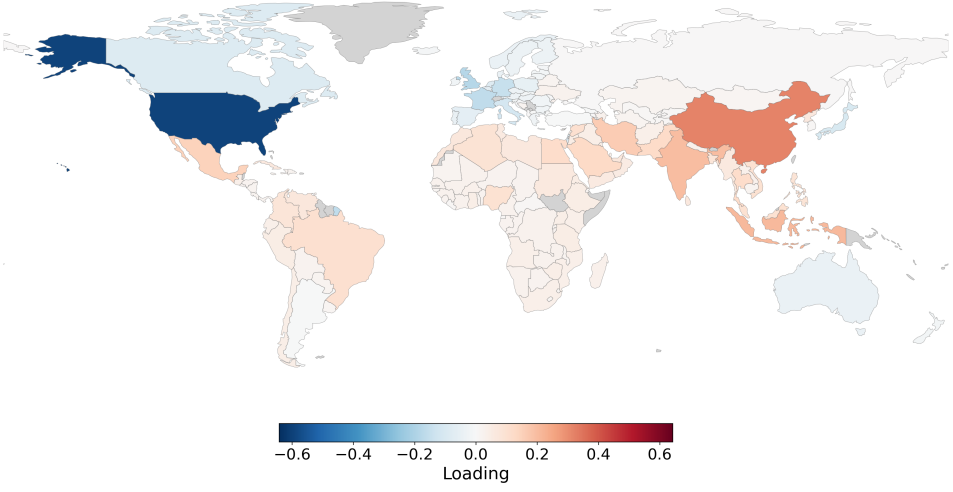
The dominant cleavage: 1970s



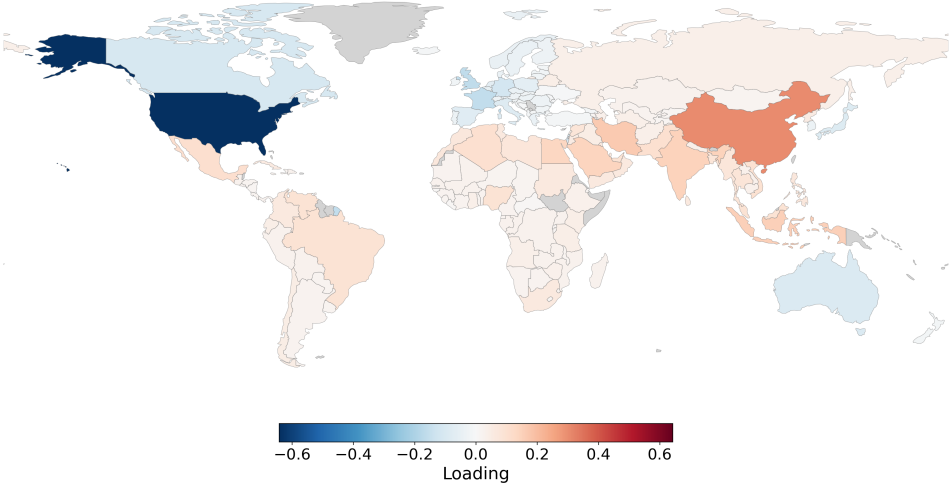
The dominant cleavage: 1980s



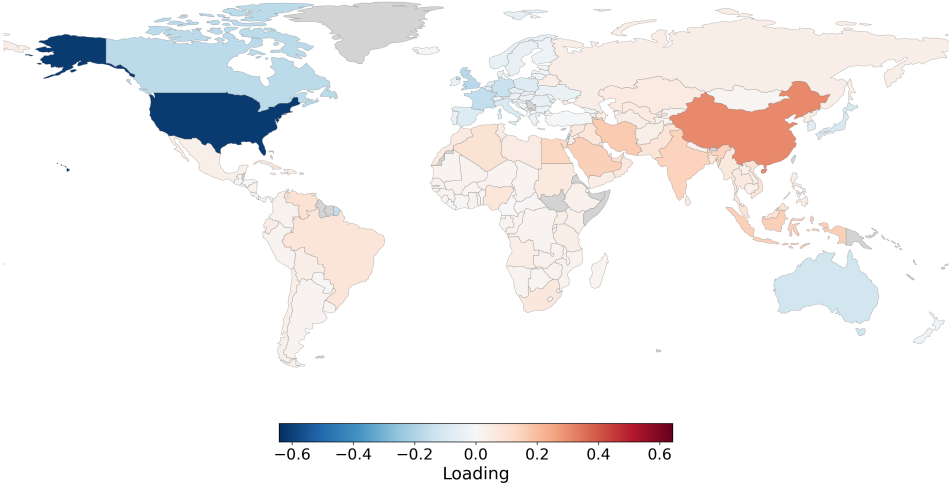
The dominant cleavage: 1990s



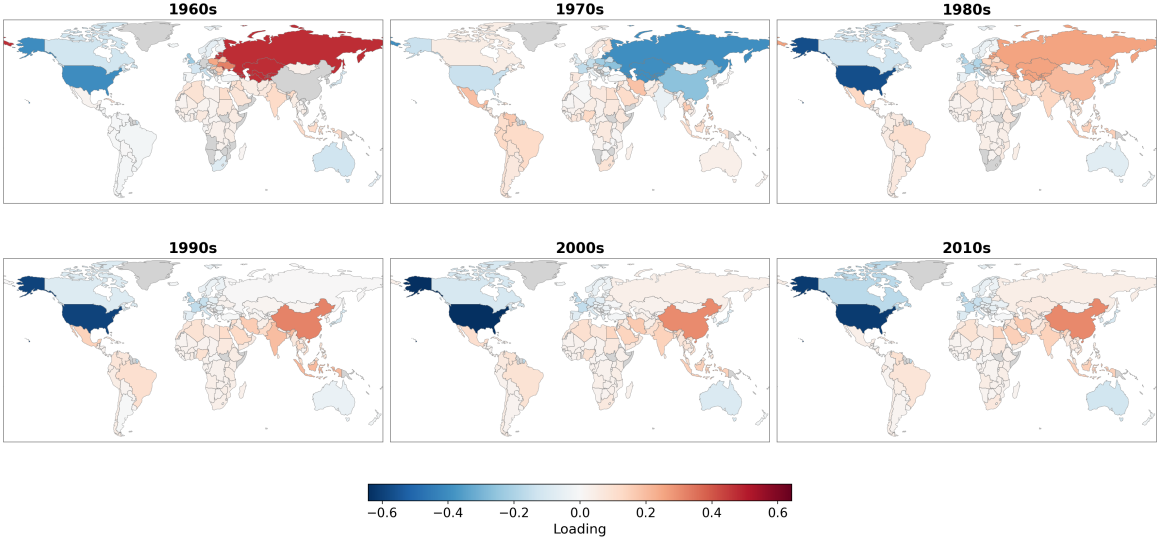
The dominant cleavage: 2000s



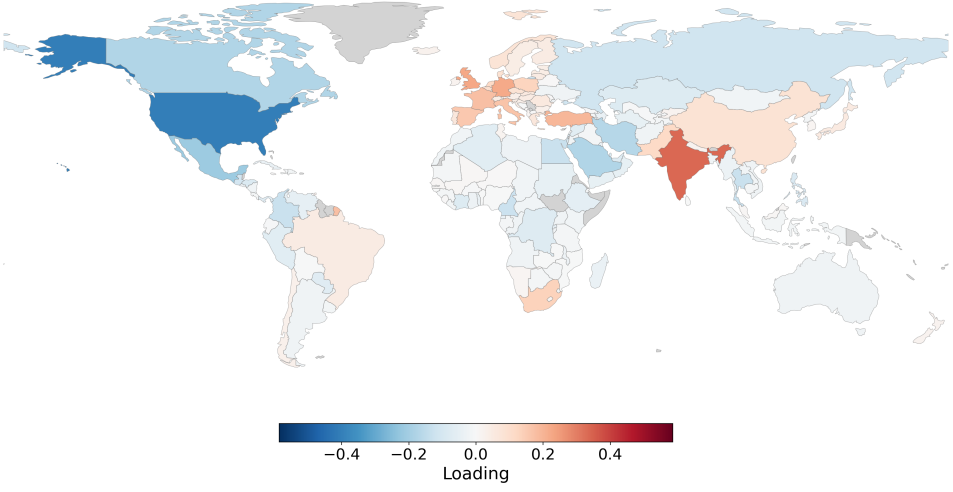
The dominant cleavage: 2010s



The dominant cleavage over time

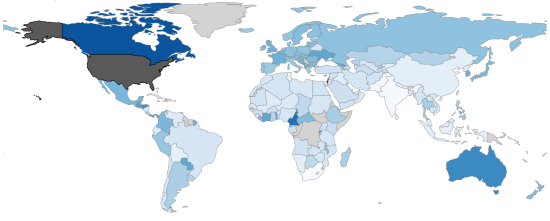


The second cleavage of the 2010s

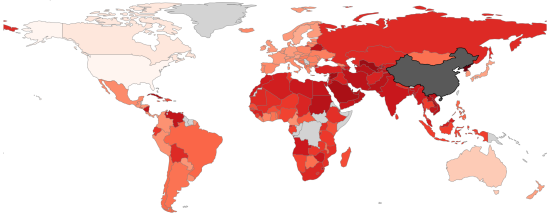


Who places high weight on the US and China? (2010s)

$G_{i,USA} - \bar{G}$: importance of USA to countries i

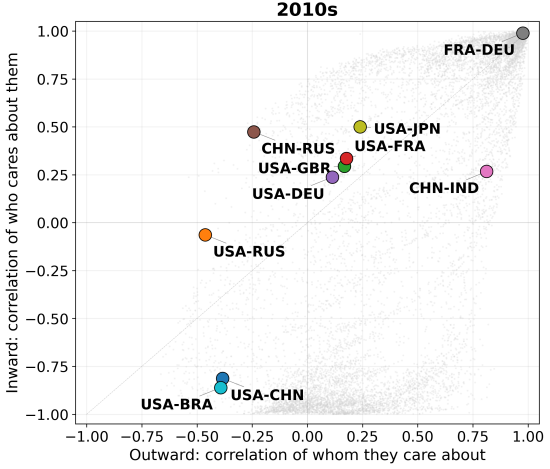


$G_{i,CHN} - \bar{G}$: importance of CHN to countries i



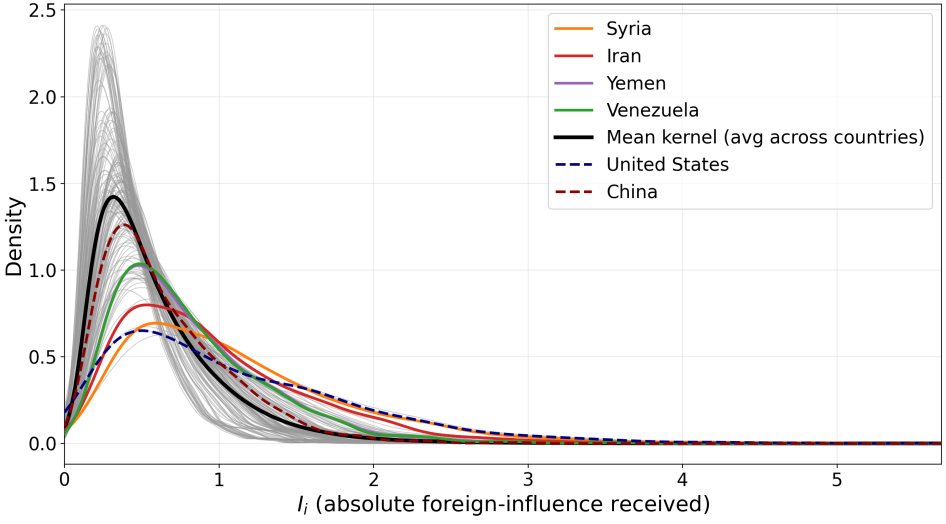
Heterogeneity in countries' networks (2010s)

- Each dot is a country pair: similarity in how much they care about others (outward, x) and how much others care about them (inward, y)
- US and China: opposite networks in both directions; France and Germany: nearly identical; China and Russia: opposite outward, similar inward



Which countries receive the most influence in the 2010s?

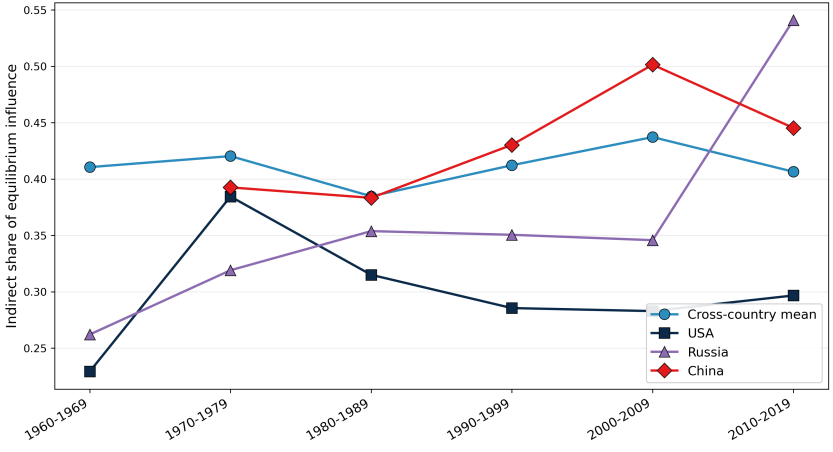
- Draw issue $\boldsymbol{\theta}_k \sim N(\mathbf{0}, \boldsymbol{\Sigma}_\theta)$, compute implied influence $|\Omega|_k$, then aggregate across issue draws k



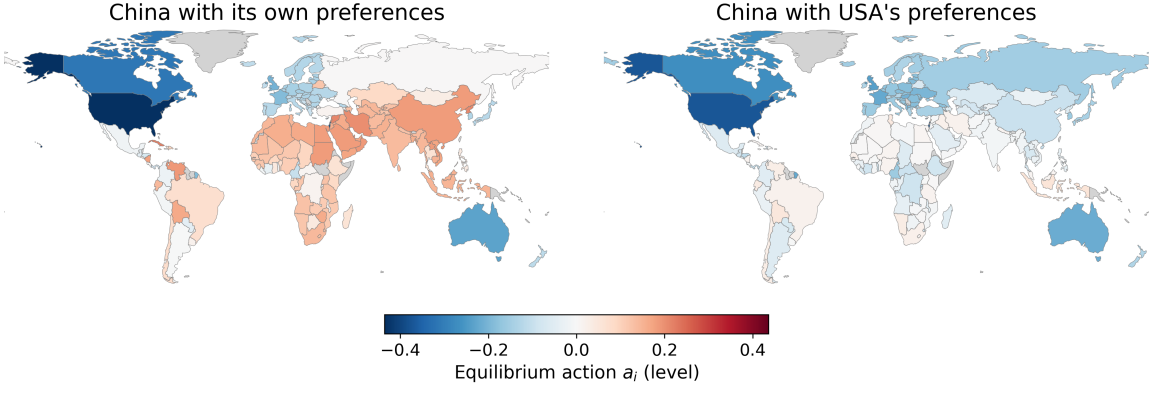
About 40% of influence runs through indirect connections

- For all $i \neq k$,

$$\kappa(\pi_k/\pi_i)\omega_{ki} = \underbrace{G_{ik}(a_i - a_k)}_{\text{direct}} + \underbrace{\sum_j G_{ij}(a_i - a_j)G_{jk} + \sum_j G_{ij}(a_i - a_j)(G^2)_{jk} + \ldots}_{\text{indirect}}$$



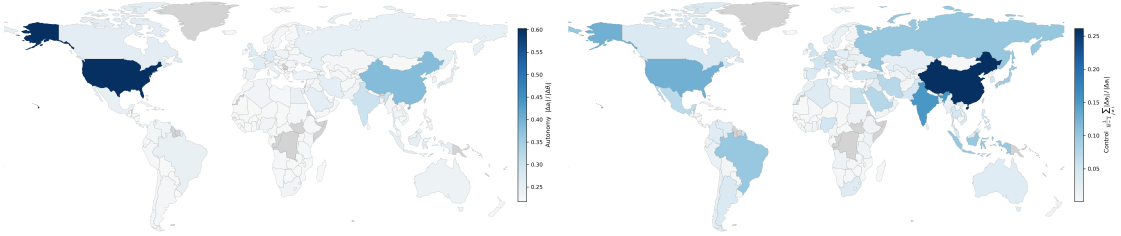
Counterfactual: China with US interests (dominant cleavage)



Counterfactuals: three notions of geopolitical power

- Autonomy: does a country's action track its own interests? $|\Delta a_i| \,/\, |\Delta \theta_i|$
- Control: how much do others move when it moves? $\frac{1}{N-1} \sum_{j \neq i} |\Delta a_j| \,/\, |\Delta a_i|$
- Shaping the game: how much does the equilibrium change if the country is removed?
  - $\mathbb{E}\left[\| \boldsymbol{a}_{-i}(\boldsymbol{G}_{-i}) - \boldsymbol{a}_{-i}(\boldsymbol{G}) \|\right]$, expectation over 1,000 international issues $\boldsymbol{\theta} \sim N(\mathbf{0}, \boldsymbol{\Sigma}_\theta)$ for 2010–2019

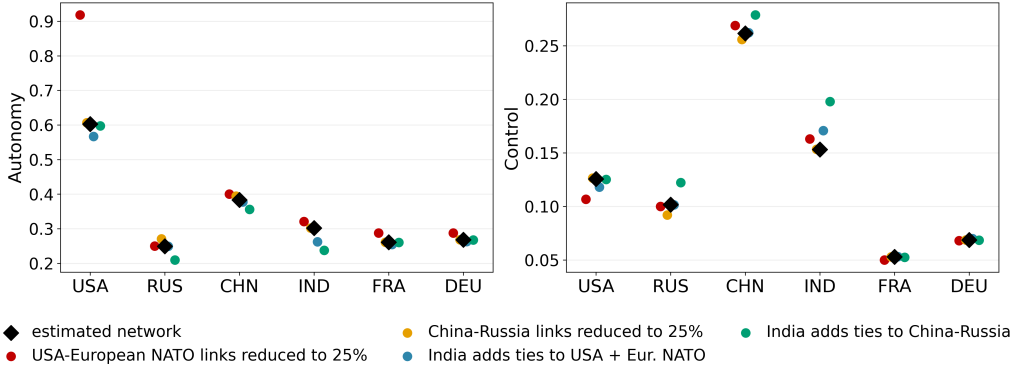
Autonomy and control across countries (2010s)



- Autonomy (left): US 0.60, China 0.38, India 0.30; the median country 0.22
  - about 80% of an interest shift is absorbed by coordination and influence
- Control (right): China 0.26, India 0.15, US 0.13
  - other countries place twice the weight on China as on the US

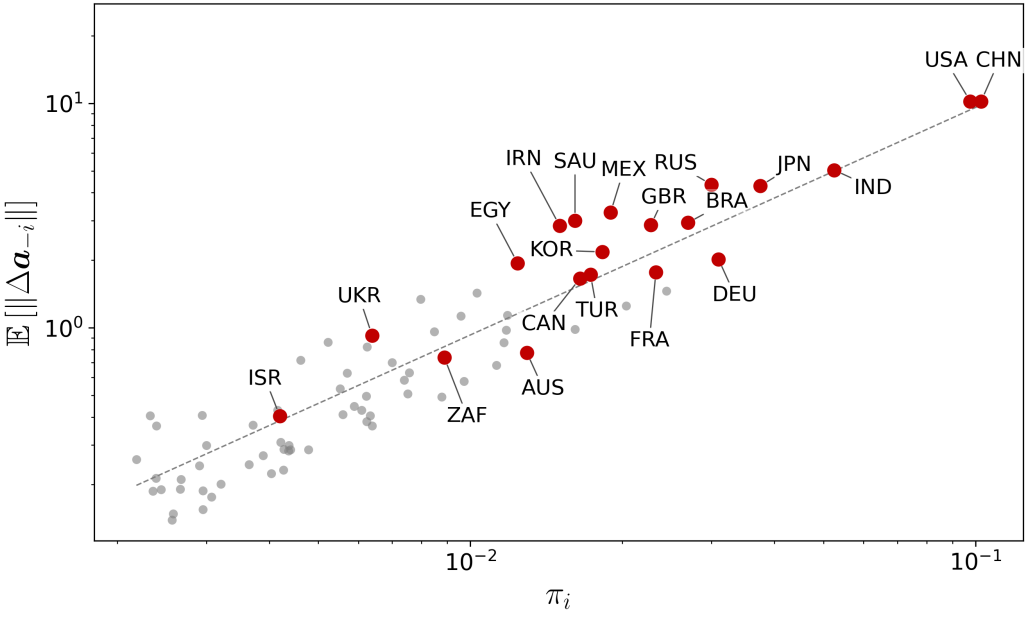
Counterfactuals for network connections

- Weaken US–European NATO or China–Russia ties by 75%; India adds ties to either bloc



- Weakening US–Europe ties raises US autonomy (0.60 $\rightarrow$ 0.92) but lowers its control
  - alliances pull the US toward allies' interests yet amplify its reach
- Tilting either way costs India autonomy but gains it control, especially toward China–Russia

Ability to shape the game: $\mathbb{E}\left[\left\|\boldsymbol{a}_{-i}\left(\boldsymbol{G}_{-i}\right)-\boldsymbol{a}_{-i}\left(\boldsymbol{G}\right)\right\|\right]$



Which shocks matter most?

Counterfactual exercise	s	ε_a	UN votes, 2010s: Votes	UN votes, 2010s: Resolutions
Changes in Domestic Interests				
China with US interests	0.209	2.35	9,726	289
China IQR preference change	0.190	2.35	6,088	253
US IQR preference change	0.143	2.20	3,408	533
India IQR preference change	0.101	1.12	249	34
Changes in Network Connections				
Retain 25% of US–European NATO links	0.177	0.75	284	199
India strengthens US–European NATO links	0.177	0.41	0	0
India strengthens China–Russia links	0.177	0.17	0	0
Retain 25% of China–Russia links	0.040	0.28	0	0
Exclude China from the network	0.543	0.54	1,090	65
Exclude the US from the network	0.540	0.56	600	352

Part of a broader research agenda on geoeconomics

- Kleinman, Liu, and Redding (2024), "International Friends and Enemies," *American Economic Journal: Macroeconomics*
- Liu and Yang (forthcoming), "International Power," *Quarterly Journal of Economics*
- More in progress...

Robustness: permutation placebo and residual bootstrap

- Left: reshuffle countries in $\boldsymbol{\Sigma}_a$ before recovering $\boldsymbol{G}$ $\Longrightarrow$ coefficients shrink toward zero
- Right: redraw the residual beyond rank 5 $\Longrightarrow$ coefficients barely move

